Science practice booklet: Grade 7 expectations

Grade 7 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Science at Grade 7 level, following the Next Generation Science Standards standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Living things

Standards practised: MS-LS3-1, MS-LS3-2, MS-LS4-4, MS-LS4-5

A worked example

A group of desert beetles has two colours: pale sand-coloured beetles and dark beetles. Birds find the dark beetles easily against the pale sand and eat more of them. Some beetles in each colour survive and have young. What is likely to happen to the colours of the beetles over many generations, and why?

1. Colour is an inherited trait, so young beetles tend to have the colour of their parents.
2. The pale beetles are hidden better, so more of them survive long enough to reproduce.
3. They pass on the genes for pale colour more often than the dark beetles pass on theirs.
4. Generation after generation, a larger share of the beetles will be pale. This is natural selection.

More and more of the beetles will be pale, because pale beetles survive and reproduce more often and pass on their genes.

Questions

1. What is a gene, and where are genes found?

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2. Which of these is an inherited trait?

   A. Riding a bicycle

   B. Speaking Arabic

   C. The shape of a seed

   D. Knowing the way to school

3. A falcon chick hatches at a falcon centre in the UAE. How many parents gave genes to the chick, and why is the chick not an exact copy of either parent?

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4. What is asexual reproduction? Give one example from plants.

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5. Date palms are either male trees or female trees. Pollen from a male flower must reach a female flower before dates can form. What is this process called, and which type of reproduction is it?

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6. Salem and his sister have the same parents, but they do not look exactly alike. Which statement explains this?

   A. Salem got all of his genes from his father

   B. Salem's genes changed because he is older

   C. They eat different foods at home

   D. Each child gets a different mixture of genes from the same two parents

7. What is a mutation? Can a mutation be harmful, helpful or have no effect?

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8. The Arabian oryx has a pale coat that helps reflect sunlight. Why is the pale coat called an adaptation?

   A. It is an inherited feature that helps the oryx survive in the hot desert

   B. The oryx grew it by standing in the Sun

   C. It changes colour every day to match the sand

   D. Every animal in every habitat has this feature

9. In a group of hares, some have longer legs and can run faster from a fox. What is likely to happen over many generations?

   A. Every hare will grow longer legs by running more

   B. More of the hares in the group will have long legs

   C. Hares with short legs will grow longer legs

   D. Leg length will stay the same because traits are not passed on

10. Fossils show that many living things that lived long ago look different from living things today. What does this suggest about living things over a very long time?

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11. A farmer plants seeds only from the tomato plants that gave the biggest fruit, year after year. What is this called?

    A. Natural selection by predators

    B. A mutation made by the farmer

    C. Selective breeding

    D. Asexual reproduction

12. A long dry spell leaves very little plant food in a desert reserve. Only some of the gazelles in a herd survive. Using the ideas of variation and natural selection, explain why.

Section 2: Earth and space

Standards practised: MS-ESS2-5, MS-ESS2-4, MS-ESS2-6, MS-ESS3-3, MS-ESS3-4, MS-ESS3-5

A worked example

The table gives the average yearly temperature in a city (example data): 1995: 26.8 C, 2005: 27.3 C, 2015: 27.8 C, 2025: 28.4 C. How much did the average temperature rise from 1995 to 2025, and what does this pattern suggest?

1. Subtract the first value from the last: 28.4 - 26.8 = 1.6.
2. Check the pattern: each reading is higher than the one before it.
3. A steady rise over thirty years is a trend, not a single hot year.
4. Scientists look for this kind of long-term trend, across many places, as evidence of a changing climate.

It rose by 1.6 C, a steady upward trend over 30 years.

Questions

1. Which of these is a non-renewable resource?

   A. Sunlight

   B. Wind

   C. Moving water in a river

   D. Natural gas

2. Why is a solar park a good choice for electricity in the UAE?

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3. Burning petrol releases carbon dioxide. Why does this gas matter for the climate?

   A. It cools the air around Earth

   B. It is a greenhouse gas that traps heat in the atmosphere

   C. It blocks all sunlight from reaching Earth

   D. It is the gas that plants give out when they make food

4. Name two pieces of evidence that Earth's climate has been warming in recent decades.

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5. Which statement describes climate rather than weather?

   A. Summers in Abu Dhabi are usually hot and humid

   B. It is cloudy in Sharjah this morning

   C. The temperature in Dubai reached 41 °C today

   D. Fog covered the road at dawn

6. On a hot afternoon at a Dubai beach, Aisha feels a cool wind blowing from the sea to the land. Warm air rises and cooler air sinks. Use this to explain the sea breeze.

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7. After a rain shower, a puddle on a school playground in Ajman disappears on a sunny day. What happened to the water, and where did the energy for this come from?

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8. Which list shows the water cycle in the correct order?

   A. Condensation, evaporation, precipitation

   B. Precipitation, condensation, evaporation

   C. Evaporation, condensation, precipitation

   D. Evaporation, precipitation, condensation

9. Desalination plants give the UAE fresh water, but they use a lot of energy. Give one way to reduce how much water we need to make.

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10. Mangrove trees take in carbon dioxide and protect the coast. Give two reasons why planting more mangroves is a good way to care for the environment.

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11. As more families move into a UAE city, what usually happens to the amount of water, food and energy the city needs? Why?

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12. Which way of getting to work usually puts the least carbon dioxide into the air for each person?

    A. Driving a car alone

    B. Taking a taxi alone

    C. Driving a larger car alone

    D. Riding a full metro train

Section 3: Matter, forces and energy

Standards practised: MS-PS1-1, MS-PS1-2, MS-PS1-5, MS-PS2-1, MS-PS2-2

A worked example

Hamdan seals a fizzy tablet and some water inside a strong bag. The bag and its contents have a mass of 150 g. The tablet fizzes and the bag swells with gas. What is the mass of the sealed bag after the reaction, and why?

1. A chemical reaction rearranges atoms into new substances.
2. No atoms are made or destroyed, and nothing can escape from a sealed bag.
3. So the total mass stays the same: it is still 150 g.
4. The gas is still inside the bag. If the bag were open, the gas would escape and the scale would show less.

150 g, because mass is conserved: the atoms are only rearranged and none left the bag.

Questions

1. What is the difference between an atom and a molecule?

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2. A water molecule is written H2O. How many atoms does one molecule have in total?

   A. 2 atoms

   B. 3 atoms

   C. 4 atoms

   D. 1 atom

3. Mariam mixes baking soda and vinegar in the kitchen and sees bubbles. Bubbles of gas are one sign that a chemical reaction may have taken place. Name one other sign.

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4. A fisherman in Fujairah leaves an iron hook on his boat, and it becomes covered in orange-brown rust. Is rust the same substance as iron? How do you know?

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5. 4 g of hydrogen react completely with 32 g of oxygen to make water. What is the mass of the water made?

   A. 36 g

   B. 28 g

   C. 32 g

   D. 128 g

6. A reaction begins with 8 atoms. How many atoms are there after the reaction? Explain.

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7. Noura rides her bicycle 36 km in 3 hours at a steady speed. What is her speed?

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8. A book rests on a table. Gravity pulls it down, yet it does not move. Why?

   A. Gravity does not act on objects at rest

   B. Friction between the book and the table cancels gravity

   C. The table pushes up with a force equal to gravity, so the forces are balanced

   D. The book is too light for gravity to act on

9. A ball rolls across a classroom floor and gradually stops. Which force makes it stop, and are the forces on it balanced?

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10. A bus brakes suddenly and the passengers lean forward. Which statement explains this?

    A. The brakes push the passengers forward

    B. The bus pulls the passengers forward

    C. Gravity pulls the passengers forward

    D. The passengers' bodies tend to keep moving forward as the bus slows

11. Khalid pushes an empty shopping trolley and then pushes the same trolley full of shopping, using the same force. Which one speeds up more, and why?

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12. A rocket pushes hot gas downwards and the rocket moves upwards. Which law does this show?

    A. Newton's third law

    B. Newton's first law

    C. Newton's second law

    D. The law of conservation of mass

Answer guide for parents

Section 1: Living things

1. A gene is a section of DNA that carries the instructions for one trait, like a recipe for one feature. Genes sit on chromosomes inside the nucleus of the cell.
2. C. The shape of a seed. is passed on through genes, so it is inherited. If your child chose B, they forgot that a language is learned from the people around us.
3. Two parents. Each gives half of the genes, half from the mother and half from the father. These genes mix in a new way, so the chick is not a copy of either parent.
4. Only one parent is needed, so no genes are mixed and the offspring have the same genes as the parent. Example: a new date palm grows from an offshoot of the parent palm.
5. Pollen moves from the male flower to the female flower, which is pollination. It is sexual reproduction because genes from two parents join in the seed.
6. D. Each child gets a different mixture of genes from the same two parents. so brothers and sisters vary. If your child chose C, they thought food causes the difference, but looks mostly come from genes.
7. A mutation is a change in a gene. It can be harmful, helpful or make no difference, depending on what the change does in that environment.
8. A. It is an inherited feature that helps the oryx survive in the hot desert. A pale coat is inherited and helps the oryx survive the heat, which is what an adaptation is. If your child chose B, they thought animals grow features during life, but adaptations are passed on in genes.
9. B. More of the hares in the group will have long legs. Faster hares escape the fox, live longer and have more young, so more hares inherit long legs. If your child chose A, they thought running makes legs grow, but traits come from the parents' genes.
10. Fossils of old living things differ from today's, so living things must have changed slowly over a very long time. Many kinds that lived then are extinct now.
11. C. Selective breeding. People pick which plants pass on their seeds each year, so the fruit tends to get bigger. If your child chose A, they mixed it up with nature doing the choosing, but here the farmer chooses.
12. The herd varies, so some gazelles need less food or water and survive the dry spell. They live to have young and pass on those traits, which is natural selection.

Section 2: Earth and space

1. D. Natural gas. took millions of years to form, so it can run out. If your child chose B, they forgot that wind is replaced all the time and never runs out.
2. The UAE has lots of strong sunshine, sunlight never runs out, and solar panels make electricity without burning fuel. So a solar park gives a lot of clean power.
3. B. It is a greenhouse gas that traps heat in the atmosphere. Carbon dioxide is a greenhouse gas that traps heat, so more of it warms Earth. If your child chose A, they thought it cools the air, but it holds heat in.
4. Any two of these show warming: rising global average temperatures, melting glaciers and ice sheets, rising sea levels, warmer oceans. Each one is measured over many years.

5. A. Summers in Abu Dhabi are usually hot and humid. Climate is the usual pattern over many years, and usually hot and humid describes that. If your child chose C, they picked one day's temperature, which is weather.
6. The land heats faster than the sea, so the air over land warms and rises. Cooler air from the sea moves in to take its place, and that moving air is the sea breeze.
7. The sunny day warmed the water until it evaporated, changing from liquid to water vapour in the air. The energy came from the Sun.
8. C. Evaporation, condensation, precipitation. Water evaporates first, then the vapour cools and condenses into clouds, then it falls as precipitation. If your child chose B, they started with rain, but rain falls only after clouds form.
9. Using less water means less to make. For example: fix leaks, turn the tap off while brushing teeth, or reuse treated water for irrigation. Any sensible idea that saves water is fine.
10. Mangroves take carbon dioxide from the air, shelter fish and birds, and protect the shore from waves. Any two of these is a good answer, because each helps the environment.
11. The need grows, because more people use water, food and energy every day. Twice as many people use about twice as much.
12. D. Riding a full metro train. Many people share one train, so the fuel or electricity used for each person is small. If your child chose A, they forgot that a car alone carries only one person for the same fuel.

Section 3: Matter, forces and energy

1. An atom is one very small particle of matter. A molecule is made when two or more atoms join together, so it is bigger.
2. B. 3 atoms. H_2 means 2 hydrogen atoms and O means 1 oxygen atom, so 2 + 1 = 3. If your child chose A, they counted only the 2 and forgot the oxygen atom.
3. Look for a new substance forming. Any one of these is a good answer: a colour change, a new solid forms, or heat or light is given out.
4. No. Iron reacts with oxygen and water to make rust, a new substance with a different colour and hardness, so a chemical reaction has happened.
5. A. 36 g. Mass is conserved in a reaction, so the water has the same mass as what went in: 4 + 32 = 36 g. If your child chose C, they picked the mass of the oxygen alone.
6. 8 atoms. Atoms only rearrange into new substances, and none are made or destroyed, so the count stays the same.
7. Speed = distance ÷ time = 36 ÷ 3 = 12 km per hour.
8. C. The table pushes up with a force equal to gravity, so the forces are balanced. The table pushes up as hard as gravity pulls down, so the forces balance and the book stays still. If your child chose A, they thought gravity stops acting on things at rest, but it always pulls.
9. Friction between the ball and the floor slows it. The forces are not balanced, because a ball with balanced forces would keep rolling at the same speed.
10. D. The passengers' bodies tend to keep moving forward as the bus slows. The bus slows but the passengers' bodies tend to keep moving forward, as Newton's first law says. If your child chose A, they thought the brakes push people forward, but the brakes act on the bus.
11. The empty trolley. With the same force, less mass means more acceleration, so the empty one speeds up more.

12. A. Newton's third law. The rocket pushes gas down and the gas pushes the rocket up with an equal force, which is Newton's third law. If your child chose B, they picked the law about things keeping their motion, not pairs of forces.